

MODULE 7: TESTING AND DEBUGGING

QA CHECKLIST

Project: Unified Military Analytics and Comparison Dashboard

Tool Used: Power BI

A. Data Accuracy Validation

- ☐ Manpower values match the source dataset
- ☐ Aircraft counts verified against raw data
- ☐ Naval fleet values validated
- ☐ Defense budget figures verified
- ☐ Power Index Rank cross-checked with dataset
- ☐ Assets per Capita calculation verified
- ☐ Coalition totals manually validated using sample selections
- ☐ No missing, duplicated, or incorrect country values
- ☐ No unexpected blank values in KPIs or charts

B. Slicer and Filter Testing

- ☐ Country slicer updates all visuals correctly
- ☐ Compare Powers – Country A selection functions correctly
- ☐ Compare Powers – Country B selection functions independently
- ☐ Multi-select coalition slicer aggregates values properly
- ☐ Clearing slicers resets visuals appropriately
- ☐ Sync Slicers function correctly across report pages
- ☐ Cross-filtering between visuals works as expected

C. DAX Measure Validation

- ☐ SUM measures return accurate totals
- ☐ CALCULATE logic correctly isolates Country A and Country B
- ☐ Coalition aggregation logic works for multiple country selections
- ☐ Measures handle no-selection scenarios correctly
- ☐ Measures do not produce incorrect context transitions
- ☐ All KPIs dynamically update based on slicer context

D. Navigation and Integration Testing

- ☐ Navigation buttons function correctly across all dashboards
- ☐ Quick Stats → Nation Overview navigation works
- ☐ Nation Overview → Compare Powers navigation works
- ☐ Compare Powers → Coalition Builder navigation works
- ☐ Navigation maintains required filter context
- ☐ No broken page links or incorrect redirections
- ☐ All four dashboards are fully integrated

E. Visual and Layout Validation

- ☐ No mislabeled charts or KPI cards
- ☐ All chart titles match displayed metrics
- ☐ Units formatted correctly (Thousands, Millions, Billions)
- ☐ Consistent number formatting across pages
- ☐ No overlapping visuals
- ☐ Proper alignment and spacing maintained
- ☐ Consistent theme, colors, and font usage
- ☐ No unnecessary scrollbars inside visuals

F. Performance and Model Optimization

- ☐ Report loads within acceptable time
- ☐ Slicers respond without noticeable delay
- ☐ No redundant DAX measures
- ☐ Relationships configured correctly (one-to-many where required)
- ☐ Cross-filter direction verified
- ☐ Star-schema model structure maintained
- ☐ File size optimized and cleaned

G. Edge Case and Scenario Testing

- ☐ Tested with highest-ranked country
- ☐ Tested with lowest-ranked country
- ☐ Tested with same country selected in Compare Powers
- ☐ Tested with no country selected
- ☐ Tested coalition with single country
- ☐ Tested coalition with multiple countries

Final Review Confirmation

- ☐ All dashboards fully functional
- ☐ Data integrity validated
- ☐ No visual or calculation errors detected
- ☐ Final file saved as: global_military_firepower_2025.pbix
- ☐ Report ready for submission